

Non-Technical Loss Investigations Management - TGI Use Case Notes



[UC29-01] - Awesense Energy Transition Platform Use Case

Overview

This manual is intended to:

- Provide instructions to create the selected Use case in True Grid Intelligence (TGI).

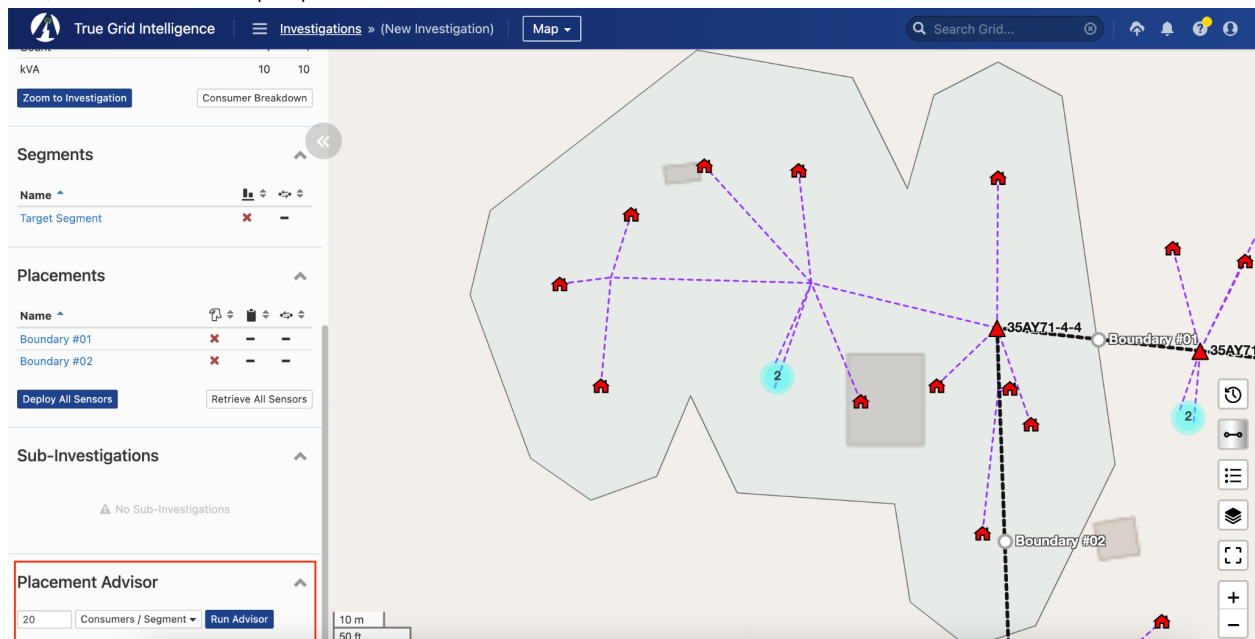
For more details about the Analysis of the Grid's Segments and Elements and how it can be analyzed using Awesense's Energy Transition Platform, please refer to the [UC29-01 - Non-Technical Loss Investigations Management.pdf](#) document.

TGI Use Case Notes

This document shows examples of using this Use Case along with a description of how to create such a Use Case on your data and screenshots directly from the TGI visualization interface.

The grid can be divided into segments using Segment Analysis - this is described in UC22. These segments can then be assigned results (the analysis factors can be defined either based on internal grid records or also on publicly available data from various trusted public sources - for example, billing irregularities, higher crime rates, customer demographics, unusual consumption patterns, type of connection, etc.) using Segment Analyzer - this is described in UC23.

This initial analysis usually divides the grid into large segments. It is necessary to create smaller segments in the newly created investigation. The "Placement Advisor", located on the bottom left, is used for this purpose.



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After selecting the rules for creating the segments into which we want to divide the grid (by the number of customers or by the number of transformers or by kVA size), we need to click on the "Run Advisor" button, which will create sub-segments.

The screenshot displays the True Grid Intelligence (TGI) interface. The top navigation bar includes the TGI logo, a menu icon, and the text "Investigations » (New Investigation)". A search bar labeled "Search Grid..." is on the right. The left sidebar contains three main sections: "Segments", "Placements", and "Sub-Investigations".

Segments Table:

Name	Count	kVA
Segment #01	1	10
Segment #02	1	10
Segment #03	1	10
Target Segment	1	10

Placements Table:

Name	Count	kVA
Boundary #01	1	10
Boundary #02	1	10
Placement #01	1	10
Placement #02	1	10

The main map area shows a grid with several segments outlined in light blue. Two specific placements are highlighted: "Placement #01" and "Placement #02". The map also shows various icons representing transformers, customers, and boundaries. A scale bar at the bottom left indicates 10 m and 50 ft.

These sub-segments can be seen in the left sidebar under Segments. Placements must be deployed with measurement sensors to create energy balances.

The segment evaluation is performed in the same way as described in UC28 on page 2. For investigations, it is possible to set different types of investigations and investigation status. This then helps for easier orientation of investigation states, filtering, etc.

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True Grid Intelligence | Investigations » Investigation of 39660 | Map

Investigation of 39660

[No Case](#) ☐ Read only

Circuit
North Central Zone

GIS Date
1 Jan, 2021 9:00 PM

Owner
[awesense-petr](#)

Starts
1 Jun, 2023 5:23 PM

Description
No Description
[Show more / Update](#)

Type
Exploratory

State
Planning

Recoverability
Unknown

Ends
☒ Ongoing

For more detailed data to analyze the investigation, there is a "Report" page where it is possible to find more data about the investigation and create your own graphs.

True Grid Intelligence | Investigations » Investigation of 39660 | Report

Investigation of 39660

Overview

Case
[No Case](#)

Circuit
North Central Zone

Owner
[awesense-petr](#) ☐ Read only

Type
Exploratory

State
Planning

Recoverability
Unknown

GIS Date
1 Jan, 2021 9:00 PM

Starts
1 Jun, 2023 5:23 PM

Ends
☒ Ongoing

Created
22 Jun, 2023 5:23 PM

Last Modified
22 Jun, 2023 5:33 PM

Description
No Description

Energy Balances

Segment	Actual		Metered		Total Discrepancy			Days (used/total)		
	Net kWh	In kWh	Out kWh	Net kWh	In kWh	Out kWh				
Target Segment	146,855	146,855	0	78,808	0	78,808	68,048	3,247	46.3	21 / 53

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General Note

This description was created for TGI version 3.10.0. For future versions of TGI, the process might be slightly different. Please contact tools@awesense.com for more information.